From the Normal Equation to PCA: Eigen-directions

ES 335 / ES 678 Machine Learning, Sem I 2026 — Prof. Udit Bhatia, Prof. Krishna Prasad Miyapuram Source: `Lectures/L05_normal_equation_eigendirections_pca.pdf` (35 slides) · Previous: Loss and projection

How to use this file. Part A builds the *new* maths Lecture 5 needs, on top of Lectures 1–4 (gradients of matrix expressions, Gram matrix $X^\top X$, PSD, rank, projection). Most of it is **eigenvalues and eigenvectors**, the **spectral theorem**, **covariance** and **Lagrange multipliers**. Part B goes through all 35 slides. The "work it out" slides are solved in this PDF; here every step is expanded and checked with numbers, and extra worked examples are added where the slide only gives symbols. Part C is practice.

What this lecture is really about: one matrix question, asked twice.

$$\underbrace{X^\top X\hat{\theta} = X^\top y}_{\text{normal equation}} \xrightarrow{X^\top X = Q\Lambda Q^\top} \underbrace{\lambda_j a_j = b_j}_{\text{one scalar equation per direction}} \qquad \Big| \qquad \underbrace{\max_{q^\top q = 1} q^\top C q}_{\text{PCA}} \xrightarrow{\text{Lagrange}} \underbrace{Cq = \lambda q}_{\text{same eigen-algebra}}$$

In **regression**, eigenvalues of $X^\top X$ say how strongly the data pin down each coefficient direction (big λ = steep, stable; small λ = flat, sensitive). In **PCA**, eigenvalues of the covariance C say how much the data *vary* along each feature direction (big λ = keep; small λ = cheap to drop).

Step-up path

You need. The normal equations, and what a matrix does to a vector.

1. Hold two pictures at once: coefficient space and observation space.
2. Find the directions a symmetric matrix only stretches, its eigenvectors.
3. Read the loss bowl through those directions: eigenvalues are curvatures.
4. Connect small eigenvalues to unstable coefficients.
5. Maximise variance along a direction, which gives PCA as an eigenvalue problem.

You can stop when you can explain why near-duplicate columns make coefficients unstable, using eigenvalues.

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Part B — Slides · All 35 slides

Part C — Practice · Jump

Part A Maths From the Ground Up

A1 Two spaces: coefficient space vs observation space

Lecture 5's main trap is mixing up **which space a vector lives in**. Keep two spaces apart:

Space	Dimension	What lives there	Examples
coefficient / feature space	$\mathbb{R}^d$ (one axis per predictor)	θ, ∇J, eigenvectors q_j of $X^\top X$, PCA directions q	$\theta = (2.2, 2.2)$
observation space	$\mathbb{R}^n$ (one axis per data row)	y, $\hat{y} = X\theta$, r, PCA scores $z = X_c q$	$y = (2, 5, 6, 9)$

$X \in \mathbb{R}^{n \times d}$ is the bridge: X maps $\mathbb{R}^d \to \mathbb{R}^n$, and $X^\top$ maps back $\mathbb{R}^n \to \mathbb{R}^d$. So:

- $X^\top r$ is a d-vector (one number per coefficient) → it can be a gradient.
- $X^\top X$ is $d \times d$ → it acts on coefficient moves.
- Xq is an n-vector → it's the *prediction pattern* produced by moving coefficients along q.

Shape check is a correctness check (L2). If a formula adds an n-vector to a d-vector, it's wrong.

A2 The gradient, one coordinate at a time

L4 A9 gave the matrix rule $\nabla_\theta \frac{1}{2}\|y - X\theta\|^2 = X^\top(X\theta - y)$. Lecture 5 re-derives it *without matrix calculus*, using only the scalar chain rule.

Write row i out: $r_i(\theta) = y_i - \sum_{k=1}^{d} x_{ik}\theta_k$. Only the $k = j$ term contains θ_j, so

$$\frac{\partial r_i}{\partial \theta_j} = -x_{ij}.$$

Chain rule on $J = \frac{1}{2}\sum_i r_i^2$:

$$\frac{\partial J}{\partial \theta_j} = \sum_i r_i \frac{\partial r_i}{\partial \theta_j} = -\sum_i x_{ij} r_i.$$

$\sum_i x_{ij} r_i$ is "column j of X, dotted with r" $= (X^\top r)_j$. Stack $j = 1..d$: $\nabla J = -X^\top r = X^\top(X\theta - y)$.

Why the transpose? The sum runs over rows i (the n direction), and the answer must be indexed by j (the d direction). Summing over the *first* index of X is exactly what $X^\top$ does.

A3 The Gram matrix $X^\top X$

Write X by columns, $X = [x_1 \ \cdots \ x_d]$, each $x_j \in \mathbb{R}^n$ = predictor j measured on all n rows. Then entry (j, k) of $X^\top X$ is a dot product of two columns:

$$(X^\top X)_{jk} = x_j^\top x_k.$$

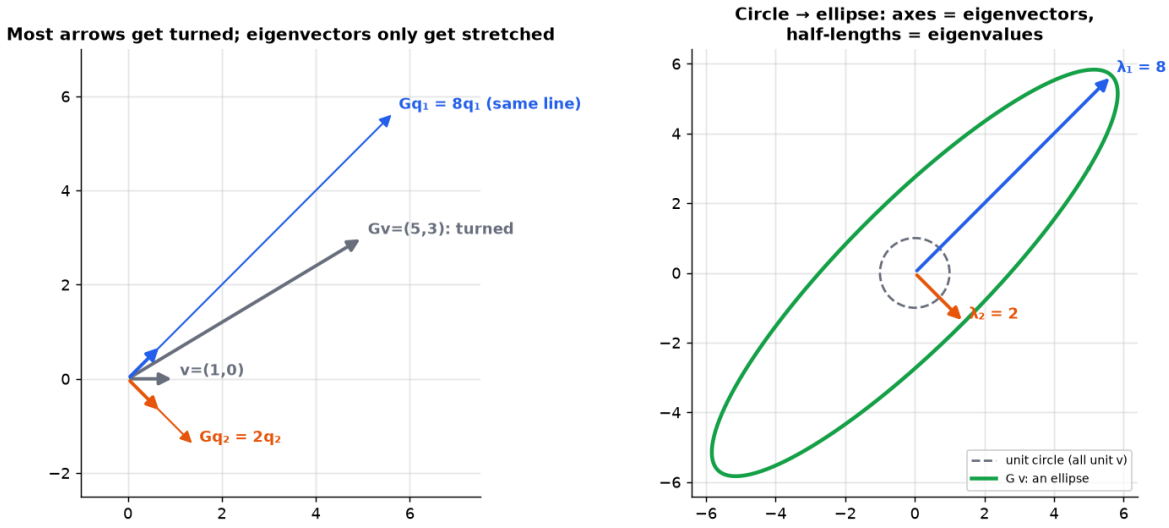
Entry	Meaning
diagonal $x_j^\top x_j = \|x_j\|^2$	the squared size of predictor j
off-diagonal $x_j^\top x_k$	how much predictors j and k **overlap** (point the same way across rows)

A matrix of pairwise dot products is called a **Gram matrix**. Properties (from L4 A10):

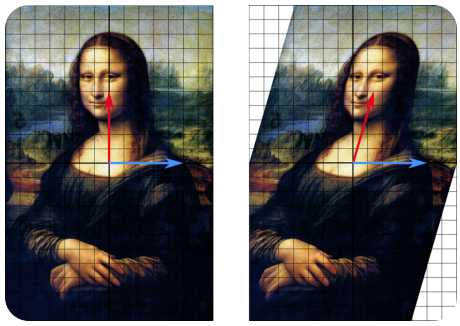
- **symmetric**: $x_j^\top x_k = x_k^\top x_j$;
- **PSD**: $v^\top X^\top X v = \|Xv\|^2 \geq 0$;
- **PD** iff full column rank.

If columns are centred, $X^\top X/(n-1)$ is the **covariance matrix** (A8): diagonal = variances, off-diagonal = covariances. Strongly correlated predictors ⇒ large off-diagonal entries ⇒ "coupling".

A4 Eigenvalues and eigenvectors



G = [[5,3],[3,5]]: (1,0) gets turned, q_1 and q_2 only stretch (×8, ×2). The unit circle becomes an ellipse along them.



A shear moves the red arrow off its line; the blue arrow (an eigenvector) stays on its line. (Wikimedia Commons: Mona_Lisa_eigenvector_grid.png, CC0)

Most matrices **rotate and stretch** a vector. A few special directions only get **stretched**:

$$Gq = \lambda q, \qquad q \neq 0.$$

q is an **eigenvector** (a direction the matrix keeps on the same line), λ its **eigenvalue** (the stretch factor; negative means flipped, zero means squashed to 0).

How to find them. $Gq = \lambda q \iff (G - \lambda I)q = 0$. A nonzero q solving this exists only if $G - \lambda I$ is singular (det = 0, it flattens the plane; see the matrix toolkit in L04 §A7):

$$\det(G - \lambda I) = 0 \qquad (\text{characteristic equation}).$$

Solve for λ, then for each λ solve $(G - \lambda I)q = 0$ for q.

2×2 shortcuts:

Fact	Formula
sum of eigenvalues	$\lambda_1 + \lambda_2 = \text{trace}(G) = a + d$
product of eigenvalues	$\lambda_1 \lambda_2 = \det G = ad - bc$
so	$\lambda = \frac{T}{2} \pm \sqrt{\frac{T^2}{4} - D}$ with T = trace, D = det
symmetric $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$	$\lambda = a \pm b$, eigenvectors $\frac{1}{\sqrt{2}}(1,1)$ and $\frac{1}{\sqrt{2}}(1,-1)$

Example. $G = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$: $\lambda = 5 \pm 3 = 8, 2$. Check $q = (1,1)$: $Gq = (8,8) = 8q$ ✓. $q = (1,-1)$: $Gq = (2,-2) = 2q$ ✓.

Scaling. If q is an eigenvector, so is cq for any $c \neq 0$. We usually pick the **unit** one: $\|q\| = 1$. (Still a sign choice: q and $-q$ are both fine.)

A5 Orthonormal vectors, orthogonal matrices, change of basis

Vectors $q_1, \ldots, q_d$ are **orthonormal** if $q_j^\top q_k = 1$ when $j = k$ and 0 otherwise (unit length, mutually perpendicular). Stack them as columns of $Q = [q_1 \ \cdots \ q_d]$. Then

$$Q^\top Q = I \quad \text{and (square } Q) \ QQ^\top = I, \qquad \text{so } Q^{-1} = Q^\top.$$

Such a Q is an **orthogonal matrix**: a rotation (possibly with a reflection). It preserves lengths: $\|Qa\| = \|a\|$.

Change of basis. Any $\theta \in \mathbb{R}^d$ can be written in the new axes $q_1..q_d$:

$$a = Q^\top \theta \quad (a_j = q_j^\top \theta, \text{ the shadow of } \theta \text{ on } q_j), \qquad \theta = Qa = a_1 q_1 + \cdots + a_d q_d.$$

The arrow θ doesn't move. Only the ruler you measure it with changes.

Example. $q_1 = \frac{1}{\sqrt{2}}(1,1)$, $q_2 = \frac{1}{\sqrt{2}}(1,-1)$, $\theta = (3,1)$. Then $a_1 = \frac{4}{\sqrt{2}} = 2\sqrt{2}$, $a_2 = \frac{2}{\sqrt{2}} = \sqrt{2}$. Rebuild: $2\sqrt{2} \cdot \frac{1}{\sqrt{2}}(1,1) + \sqrt{2} \cdot \frac{1}{\sqrt{2}}(1,-1) = (2,2) + (1,-1) = (3,1)$ ✓.

A6 The spectral theorem and quadratic forms

Spectral theorem. Every **real symmetric** $G \in \mathbb{R}^{d \times d}$ has

1. all eigenvalues **real**;
2. a full set of d **orthonormal** eigenvectors (eigenvectors for different eigenvalues are automatically perpendicular);
3. the decomposition

$$G = Q\Lambda Q^\top = \sum_{j=1}^{d} \lambda_j q_j q_j^\top, \qquad \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d).$$

Read $Gv = Q\Lambda Q^\top v$ right to left: $Q^\top$ **measures** v in eigen-coordinates → Λ **scales** each coordinate independently → Q **returns** to the original axes. In the eigenbasis, G is just a diagonal stretch.

If G is also PSD (like $X^\top X$ or C), all $\lambda_j \geq 0$. If PD, all $\lambda_j > 0$. Number of nonzero eigenvalues $= \mathrm{rank}(G)$.

Quadratic forms in the eigenbasis. With $v = Qz$:

$$v^\top G v = z^\top Q^\top Q \Lambda Q^\top Q z = z^\top \Lambda z = \sum_j \lambda_j z_j^2.$$

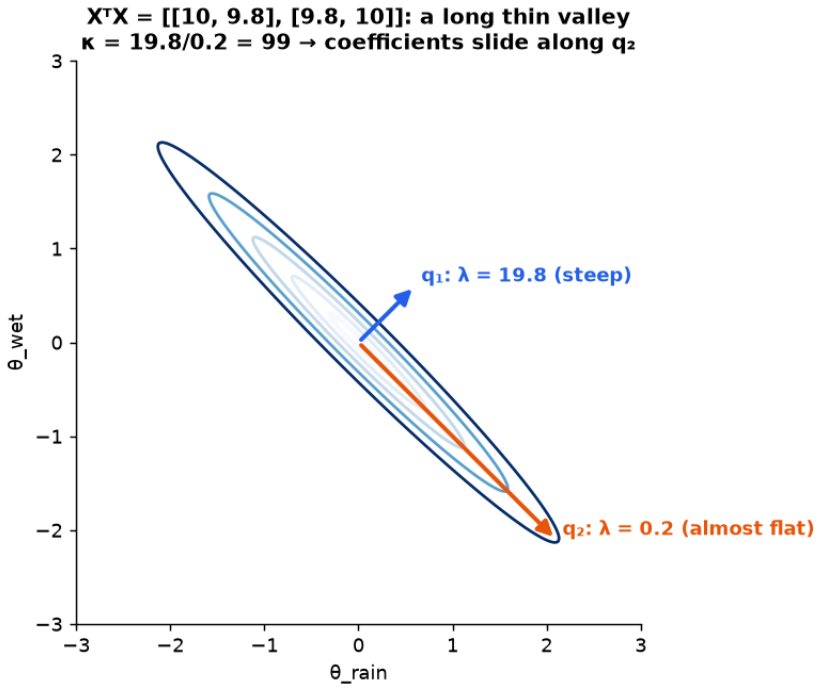
A tilted, coupled bowl becomes a sum of independent 1-D parabolas.

Rayleigh bound. For a unit vector v ($\sum z_j^2 = 1$):

$$\lambda_{\min} \leq v^\top G v = \sum_j \lambda_j z_j^2 \leq \lambda_{\max},$$

because a weighted average of the λ_j's (weights z_j^2 summing to 1) lies between the smallest and largest. **Equality at the top** when $v = q_{\max}$. This is the whole of PCA (slide 32).

A7 Condition number and sensitivity



Slide 14's matrix: steep across the valley (λ = 19.8), nearly flat along it (λ = 0.2), so the data can't pin θ down along q_2.

For a PD matrix, the **condition number** is

$$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \quad (\geq 1).$$

$\kappa \approx 1$: round bowl, every direction equally well determined. $\kappa \gg 1$: long thin valley.

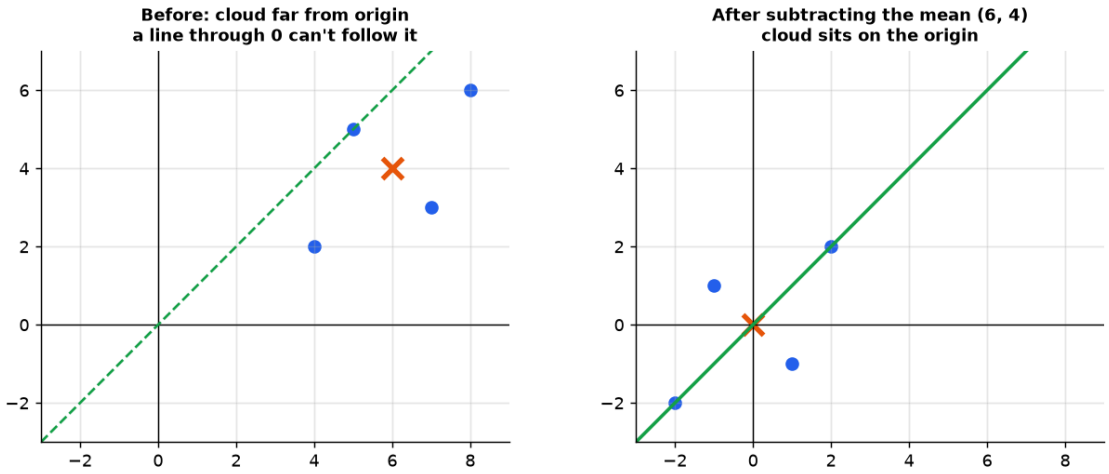
Two consequences in regression (slides 22–23):

Small λ_j means	Why
coefficient **sensitive** along q_j	$a_j = b_j/\lambda_j$: a small change in b_j gets divided by a small number
loss **flat** along q_j	curvature along q_j is λ_j
gradient descent **slow** along q_j	the gradient component is $\lambda_j z_j$: tiny push
predictions **barely change**	$\|Xq_j\|^2 = \lambda_j$: moving along q_j hardly moves $\hat{y}$

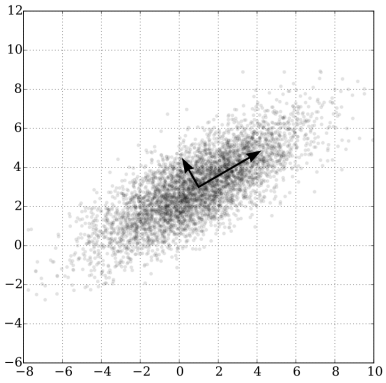
Examples. $\begin{bmatrix} 10 & 9.8 \\ 9.8 & 10 \end{bmatrix}$: $\lambda = 19.8,\ 0.2$, $\kappa = 99$. L4A's $\begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}$: $\lambda = 9 \pm \sqrt{61} \approx 16.81,\ 1.19$, $\kappa \approx 14.1$.

$\lambda_{\min} = 0$ is the extreme case: rank deficient, $\kappa = \infty$, a flat valley (L4 slide 31).

A8 Centring, variance and the covariance matrix



The toy storms before and after subtracting the mean; only after centring can a line through 0 follow the cloud.



PCA on a 2-D cloud: arrows are covariance eigenvectors, scaled by √eigenvalue. (Wikimedia Commons: GaussianScatterPCA.svg, CC BY 4.0)

Centring. Subtract each feature's mean: $x_{i,c} = x_i - \bar{x}$. Afterwards every column of X_c averages to 0: $\mathbf{1}^\top X_c = 0^\top$.

Sample variance of numbers $z_1..z_n$: $\mathrm{Var}(z) = \frac{1}{n-1}\sum_i (z_i - \bar{z})^2$. If $\bar{z} = 0$ this is $\frac{z^\top z}{n-1}$.

($n-1$, not n: the **unbiased** version, since one degree of freedom was used up estimating the mean. The argmax in PCA doesn't care which one you use: it's a positive constant.)

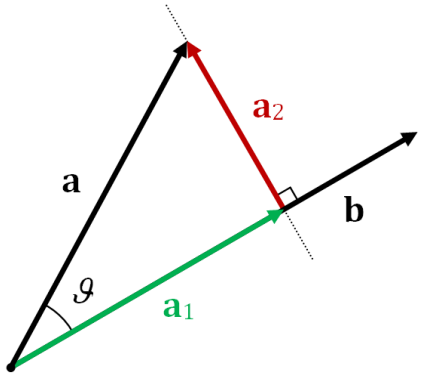
Sample covariance between features j and k: $\frac{1}{n-1}\sum_i x_{ij,c}\, x_{ik,c}$. All of them at once:

$$C = \frac{1}{n-1} X_c^\top X_c \in \mathbb{R}^{d\times d}.$$

C_{jj} = variance of feature j; C_{jk} = covariance of j and k. C is a scaled Gram matrix, so it's **symmetric PSD** (A3).

Why centre? Without it, $X^\top X/(n-1)$ measures spread around the **origin**, not around the mean. A cloud far from 0 would have a huge "variance" in the direction of its mean, even if the points barely differ.

A9 Projecting onto a single unit direction



Projecting a onto a direction b: the shadow a_1 and the perpendicular remainder a_2. (Wikimedia Commons: Projection_and_rejection.png, CC BY-SA 3.0)

Given a unit vector q ($q^\top q = 1$) and a point x, the nearest point on the line $\mathrm{span}(q)$ is zq with

$$z = q^\top x \qquad (\text{signed length of the shadow}).$$

Derivation (same perpendicular argument as L4): the leftover $x - zq$ must be perpendicular to q:

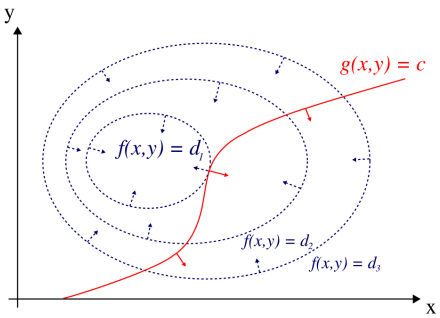
$$q^\top(x - zq) = 0 \;\Rightarrow\; q^\top x - z\underbrace{q^\top q}_{1} = 0 \;\Rightarrow\; z = q^\top x.$$

(If q weren't unit length, $z = q^\top x / q^\top q$. Fixing $q^\top q = 1$ keeps it clean.)

The projection matrix onto that line is $qq^\top$ (the 1-column case of L4's $P_X = X(X^\top X)^{-1}X^\top$ with $X = q$).

Distinguish: $q \in \mathbb{R}^d$ is a **direction** (one per dataset). $z_i \in \mathbb{R}$ is a **score** (one per observation). Stacked: $z = X_c q \in \mathbb{R}^n$.

A10 Lagrange multipliers



At the constrained optimum the contour of f is tangent to the constraint g = c, so ∇f ∥ ∇g. (Wikimedia Commons: LagrangeMultipliers2D.svg, Public domain)

Problem: maximise $f(x)$ subject to $g(x) = 0$ (you're only allowed to walk on the curve/surface $g = 0$).

Idea. At a constrained optimum you can't improve f by moving *along* the constraint. So ∇f has no component along the constraint: it points straight out of the curve, i.e. along the constraint's normal ∇g:

$$\nabla f = \lambda \nabla g \qquad \text{for some scalar } \lambda \text{ (the multiplier)}.$$

Recipe.

1. Form the **Lagrangian** $\mathcal{L}(x, \lambda) = f(x) - \lambda g(x)$.
2. Set **all** partial derivatives to zero: $\nabla_x \mathcal{L} = 0$ (gives $\nabla f = \lambda \nabla g$) and $\partial \mathcal{L} / \partial \lambda = -g = 0$ (gives back the constraint).

3. Solve. Split into cases when you divide by something that might be 0.
4. Evaluate f at **every** candidate; the largest is the max (on a closed, bounded constraint like a circle, a max is guaranteed to exist).

Mini-example. Max $x + y$ on $x^2 + y^2 = 1$. $\nabla f = (1, 1)$, $\nabla g = (2x, 2y)$. $1 = 2\lambda x$, $1 = 2\lambda y \Rightarrow x = y$. Constraint: $x = y = \pm\frac{1}{\sqrt{2}}$. Max $f = \sqrt{2}$ at $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$.

A11 Frobenius norm, trace and low-rank reconstruction

Frobenius norm of a matrix = treat it as one long vector:

$$\|A\|_F^2 = \sum_{i,j} A_{ij}^2 = \operatorname{trace}(A^\top A) = \sum_{\text{columns}} \|a_j\|^2.$$

For data matrices, it's the **total squared error over all entries**.

Encode / decode with orthonormal columns $Q_k = [q_1 \cdots q_k] \in \mathbb{R}^{d\times k}$ ($Q_k^\top Q_k = I_k$, but $Q_k Q_k^\top \neq I_d$ unless $k = d$):

Step	Formula	Shape
encode (scores)	$Z_k = X_c Q_k$	$n \times k$
decode (reconstruction)	$\hat{X}_k = Z_k Q_k^\top = X_c\, Q_k Q_k^\top$	$n \times d$

$Q_k Q_k^\top$ is the projection onto $\operatorname{span}(q_1..q_k)$. If $k = d$ it's I, and reconstruction is perfect.

The error, using the full eigenbasis $I = \sum_{j=1}^{d} q_j q_j^\top$:

$$X_c - \hat{X}_k = X_c \sum_{j>k} q_j q_j^\top, \qquad \|X_c - \hat{X}_k\|_F^2 = \sum_{j>k} \|X_c q_j\|^2 = (n-1)\sum_{j>k} \lambda_j.$$

(The cross terms vanish because the q_j are orthonormal. And $\|X_c q_j\|^2 = (n-1) q_j^\top C q_j = (n-1)\lambda_j$.)

So **the reconstruction error is exactly the sum of the discarded eigenvalues** (times $n - 1$).

Part B Slide by Slide (all 35 slides)

Running examples used below:

- **L4A regression data** (for checks): $X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$, $y = (2, 5, 6, 9)$, $X^\top X = \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}$, $X^\top y = (22, 44)$, $\hat{\theta} = (2.2, 2.2)$.
- **Coupled predictors** (slide 14): $X^\top X = \begin{bmatrix} 10 & 9.8 \\ 9.8 & 10 \end{bmatrix}$.
- **Toy storms for PCA** (added here; slides 25–35 are symbolic): 4 storms, already centred, (rainfall, wetness):

storm i	1	2	3	4
centred rainfall	2	1	−1	−2
centred wetness	2	−1	1	−2

Slide 1 Title

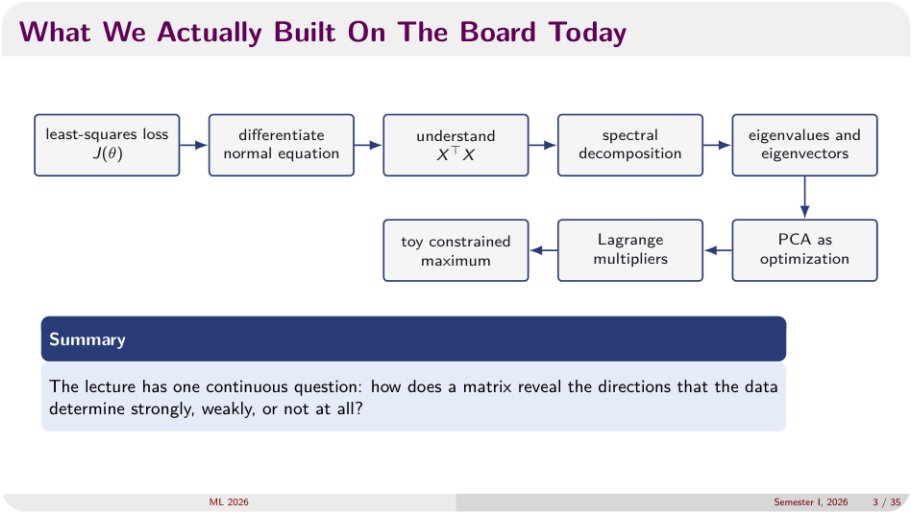
Lecture 5: From the Normal Equation to PCA. Prof. Udit Bhatia, Civil Engineering; Computer Science and Engineering, Semester I 2026.

Slide 2 Course Roadmap: You Are Here

Still in **L3–L6 estimation and linear models**. Lecture 4B established least squares and projection. Today:

1. reconstruct the normal equation carefully;
2. use **spectral decomposition** to expose its coefficient directions;
3. reuse the same eigenvalue language to formulate **PCA as constrained optimisation**.

Slide 3 What We Actually Built On The Board Today



Slide 3 — Today's chain from least squares to PCA.

least-squares loss $J(\theta)$ → differentiate → normal equation → understand $X^\top X$ → spectral decomposition → eigenvalues and eigenvectors → PCA as optimisation → Lagrange multipliers → toy constrained maximum.

> **Summary:** one continuous question: *how does a matrix reveal the directions that the data determine strongly, weakly, or not at all?*

Slide 4 Learning Objectives

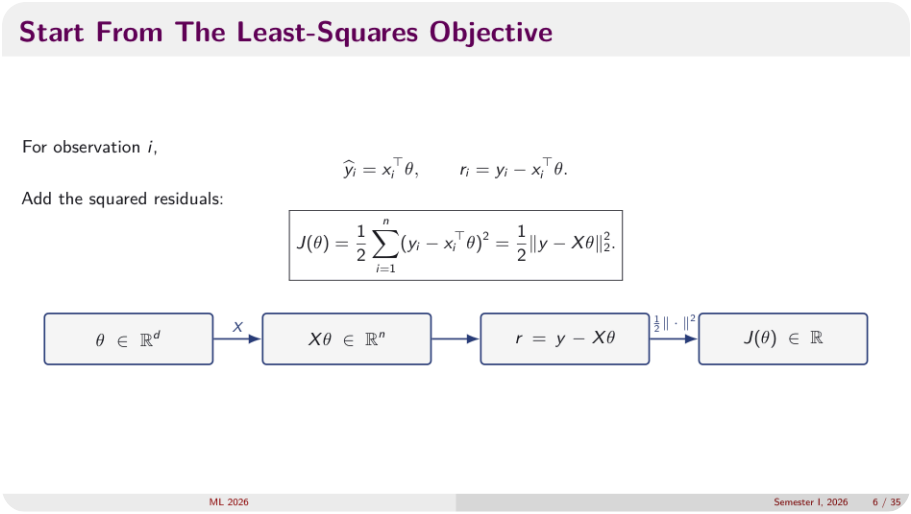
1. Derive $\nabla J(\theta) = X^\top(X\theta - y)$ without guessing the transpose (slides 8–9, A2).
2. Obtain and interpret the normal equation $X^\top X\hat{\theta} = X^\top y$ (slides 9–10).
3. Explain why $X^\top X$ is symmetric and PSD (slides 12–13).
4. Use the spectral theorem to describe regression direction by direction (slides 17–21).
5. Explain where $a = Q^\top\hat{\theta}$ and $b = Q^\top X^\top y$ come from (slides 19–20).
6. Formulate PCA by maximising projected variance (slides 27–29).
7. Derive a Lagrange-multiplier system and reuse it to obtain $Cq = \lambda q$ (slides 30–32).

Slide 5 Notation: Keep Track Of The Space Before The Algebra

Object	Shape	Meaning	Space (A1)
X	$n \times d$	n observations, d predictors	bridge
y	$n \times 1$	observed response	observation
θ	$d \times 1$	regression coefficients	coefficient
$X\theta$	$n \times 1$	fitted response	observation
$r = y - X\theta$	$n \times 1$	residual vector	observation
$X^\top X$	$d \times d$	predictor Gram matrix	acts on coefficient space
q_j	$d \times 1$	one eigenvector direction (which matrix: from context)	coefficient / feature
X_c	$n \times d$	centred feature matrix for PCA	bridge
q	$d \times 1$	one candidate PCA feature direction	feature
$z = X_c q$	$n \times 1$	projected coordinate (score) for every observation	observation
$C = \frac{1}{n-1} X_c^\top X_c$	$d \times d$	sample covariance matrix	acts on feature space

Summary: regression moves a coefficient vector θ. PCA chooses a feature direction q and produces scores $z = X_c q$. **The direction and the scores live in different spaces** ($\mathbb{R}^d$ vs $\mathbb{R}^n$).

Slide 6 Start From The Least-Squares Objective



Slide 6 — θ lives in coefficient space ($\mathbb{R}^d$); Xθ and r live in observation space ($\mathbb{R}^n$).

For observation i: $\hat{y}_i = x_i^\top \theta$, $r_i = y_i - x_i^\top \theta$ (here $x_i^\top$ is **row** i of X). Add squared residuals:

$$J(\theta) = \frac{1}{2}\sum_{i=1}^{n}(y_i - x_i^\top \theta)^2 = \frac{1}{2}\|y - X\theta\|_2^2.$$

Pipeline: $\theta \in \mathbb{R}^d \xrightarrow{X} X\theta \in \mathbb{R}^n \to r = y - X\theta \xrightarrow{\frac{1}{2}\|\cdot\|^2} J(\theta) \in \mathbb{R}$.

(The $\frac{1}{2}$ is only there to cancel the 2 from differentiating; it doesn't move the argmin.)

Slide 7 Expand The Objective Before Differentiating

$$2J(\theta) = (y - X\theta)^\top (y - X\theta) = y^\top y - y^\top X\theta - \theta^\top X^\top y + \theta^\top X^\top X\theta.$$

The two middle terms are the **same scalar** (1×1 equals its own transpose):

$$y^\top X\theta = (y^\top X\theta)^\top = \theta^\top X^\top y.$$

Therefore

$$J(\theta) = \tfrac{1}{2}y^\top y - \theta^\top X^\top y + \tfrac{1}{2}\theta^\top X^\top X\theta.$$

Summary: the transpose appears because the derivative must return a d-dimensional coefficient-space vector. ($X^\top y$ is $d \times 1$; $y^\top X$ would be $1 \times d$.)

Check on L4A data at $\theta = (2.2, 2.2)$: $y^\top y = 4 + 25 + 36 + 81 = 146$; $\theta^\top X^\top y = 2.2(22 + 44) = 145.2$; $\theta^\top X^\top X\theta = 2.2^2(4 + 6 + 6 + 14) = 4.84 \times 30 = 145.2$. $J = 73 - 145.2 + 72.6 = 0.4 = \frac{1}{2}\text{SSE}$ ✓ (SSE was 0.8).

Slide 8 Differentiate One Coefficient At A Time

Start from

$$J(\theta) = \frac{1}{2}\sum_{i=1}^{n}\left(y_i - \sum_{k=1}^{d} x_{ik}\theta_k\right)^2.$$

Let $r_i(\theta) = y_i - \sum_k x_{ik}\theta_k$. Only the $k = j$ term depends on θ_j:

$$\frac{\partial r_i}{\partial \theta_j} = -x_{ij}.$$

Chain rule ($\frac{d}{du}\frac{1}{2}u^2 = u$):

$$\frac{\partial J}{\partial \theta_j} = \frac{1}{2}\sum_{i=1}^{n} 2r_i \frac{\partial r_i}{\partial \theta_j} = -\sum_{i=1}^{n} x_{ij} r_i = \sum_{i=1}^{n} x_{ij}(x_i^\top \theta - y_i).$$

This is **component j of** $X^\top(X\theta - y)$: column j of X dotted with the vector $X\theta - y$. (A2)

Numbers (L4A, at $\theta = 0$): $X\theta - y = (-2, -5, -6, -9)$. $\partial J/\partial\theta_0 = 1 \cdot (-2 - 5 - 6 - 9) = -22$; $\partial J/\partial\theta_1 = 0(-2) + 1(-5) + 2(-6) + 3(-9) = -44$. So $\nabla J(0) = -X^\top y = (-22, -44)$ ✓: both coefficients should increase.

Slide 9 Stack The Derivatives: The Normal Equation Appears

Stack the d scalar derivatives:

$$\nabla J(\theta) = X^\top(X\theta - y) \in \mathbb{R}^d.$$

At a stationary point $\hat{\theta}$:

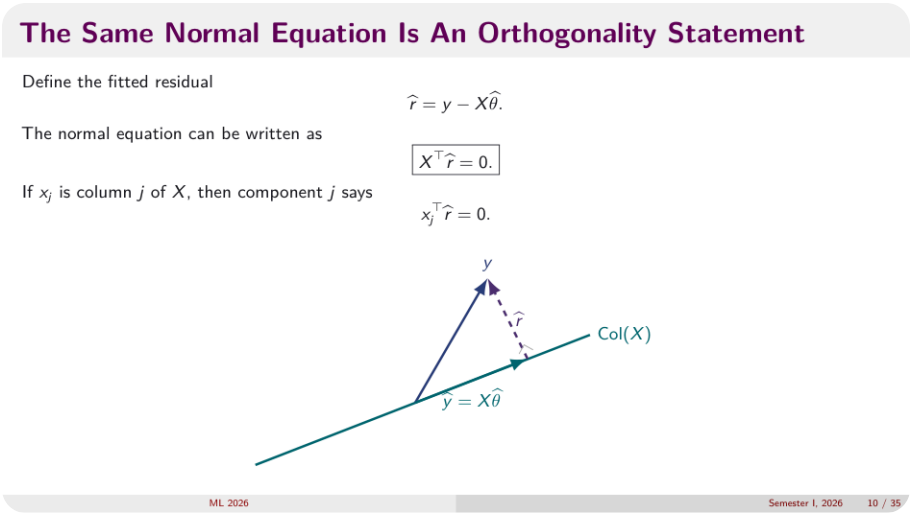
$$X^\top(X\hat{\theta} - y) = 0 \;\Rightarrow\; X^\top X\hat{\theta} - X^\top y = 0 \;\Rightarrow\; \boxed{X^\top X\hat{\theta} = X^\top y}.$$

Dimension check: $X^\top X$ is $d \times d$, $\hat{\theta}$ is $d \times 1$ → left side $d \times 1$. $X^\top$ is $d \times n$, y is $n \times 1$ → right side $d \times 1$ ✓.

Normal equation: one scalar stationarity equation **for each coefficient** (d equations, d unknowns).

Because the Hessian $X^\top X \succeq 0$ (slide 13), every stationary point is a global minimum.

Slide 10 The Same Normal Equation Is An Orthogonality Statement



Slide 10 — The residual is perpendicular to every column of X.

Define the fitted residual $\hat{r} = y - X\hat{\theta}$. The normal equation is

$$X^\top \hat{r} = 0, \qquad \text{i.e. } x_j^\top \hat{r} = 0 \;\text{ for every column } j.$$

The residual is **perpendicular to every predictor column**, hence to all of $\mathrm{Col}(X)$. Picture: y above the plane $\mathrm{Col}(X)$, $\hat{y} = X\hat{\theta}$ its foot, $\hat{r}$ the vertical drop. (L4 slide 28.)

Check (L4A): $\hat{r} = (-0.2, 0.6, -0.6, 0.2)$. Ones column: sum $= 0$ ✓. x column: $0 + 0.6 - 1.2 + 0.6 = 0$ ✓.

Two readings of one equation: calculus says "gradient is zero"; geometry says "residual $\perp$ model space".

Slide 11 When May We Write The Inverse Formula?

From $X^\top X\hat{\theta} = X^\top y$ we may write

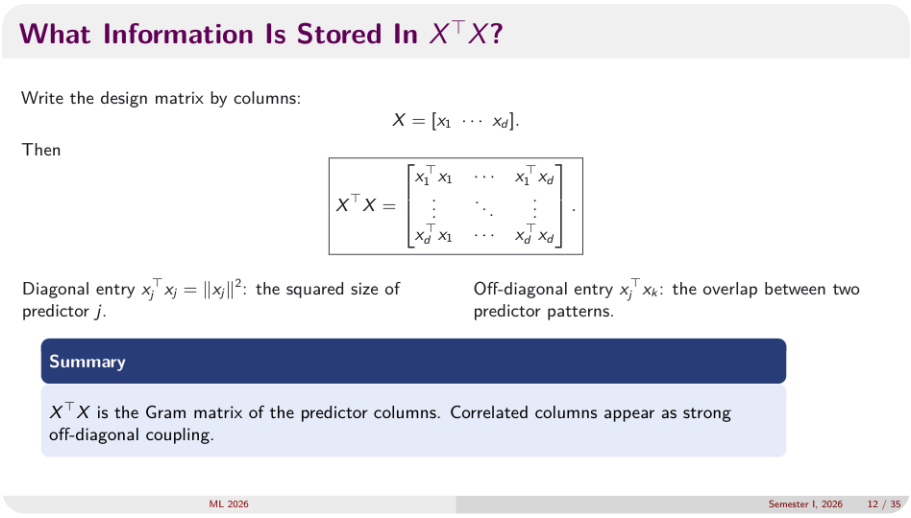
$$\hat{\theta} = (X^\top X)^{-1} X^\top y$$

only when $X^\top X$ is invertible.

Full column rank	Exact linear dependence
$\operatorname{rank}(X) = d$	some predictor is an exact combination of others
no nonzero coefficient move v satisfies $Xv = 0$	$\exists v \neq 0$ with $Xv = 0$
$X^\top X$ PD, all $\lambda_j > 0$	$X^\top X$ singular, some $\lambda_j = 0$
least-squares coefficients **unique**	normal equation still holds (has infinitely many solutions), but the **inverse formula doesn't exist**

Careful. "Solve the normal equation" is always valid. "Invert $X^\top X$" isn't. (In code, use a solver such as `np.linalg.lstsq` / `solve`, not `inv`.)

Slide 12 What Information Is Stored In $X^\top X$?



Slide 12 — Diagonal = column lengths², off-diagonal = overlaps between columns.

Write $X = [x_1 \ \cdots \ x_d]$ by columns. Then

$$X^\top X = \begin{bmatrix} x_1^\top x_1 & \cdots & x_1^\top x_d \\ \vdots & \ddots & \vdots \\ x_d^\top x_1 & \cdots & x_d^\top x_d \end{bmatrix}.$$

- **Diagonal** $x_j^\top x_j = \|x_j\|^2$: the squared size of predictor j.
- **Off-diagonal** $x_j^\top x_k$: the overlap between two predictor patterns.

> **Summary:** $X^\top X$ is the **Gram matrix** of the predictor columns. Correlated columns appear as strong off-diagonal coupling. (A3)

L4A: $\begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}$: $\|\mathbf{1}\|^2 = 4$ (= n), $\|x\|^2 = 0 + 1 + 4 + 9 = 14$, overlap $\mathbf{1}^\top x = \sum x_i = 6$.

Slide 13 Why $X^\top X$ Is Positive Semidefinite

Any coefficient direction $v \in \mathbb{R}^d$:

$$v^\top (X^\top X) v = (Xv)^\top (Xv) = \|Xv\|_2^2 \geq 0 \quad \Rightarrow \quad X^\top X \succeq 0.$$

For least squares, $\nabla^2 J(\theta) = X^\top X$, so J **is convex in** θ.

> **Summary:** equality holds exactly when $Xv = 0$: moving the coefficients along v changes **no fitted value**. Strict positivity for every nonzero v gives a **unique** minimiser.

Eigen-reading (preview): if q is a unit eigenvector, $q^\top X^\top X q = \lambda\, q^\top q = \lambda$, and that equals $\|Xq\|^2 \geq 0$. So **every eigenvalue of** $X^\top X$ **is** ≥ 0, and it measures how much moving along q changes the predictions.

Slide 14 Why Eigenvectors Enter Regression At This Point

Rainfall and wetness predictors are strongly coupled:

$$X^\top X = \begin{bmatrix} 10 & 9.8 \\ 9.8 & 10 \end{bmatrix}.$$

The original coefficient axes (θ_{rain}, θ_{wet}) mix the two effects: you can't tell them apart because they almost always move together.

Question: are there special coefficient combinations that $X^\top X$ acts on **without turning them into a different direction**? Those satisfy

$$(X^\top X)q = \lambda q.$$

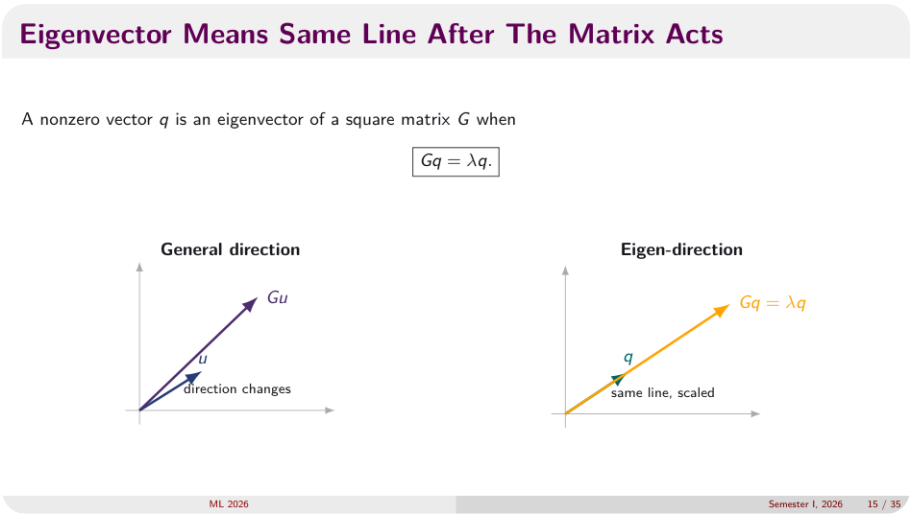
- q: a coefficient combination, e.g. "increase both coefficients together".
- λ: how strongly the data and loss react to that combination.

Answer here (A4 shortcut, $a \pm b$):

Direction	q	λ	Meaning
"both together"	$\frac{1}{\sqrt{2}}(1, 1)$	$10 + 9.8 = 19.8$	strongly determined by data
"one up, other down"	$\frac{1}{\sqrt{2}}(1, -1)$	$10 - 9.8 = 0.2$	barely determined: the data can't separate rain from wetness

Ratio $19.8/0.2 = 99$ (condition number, A7).

Slide 15 Eigenvector Means Same Line After The Matrix Acts



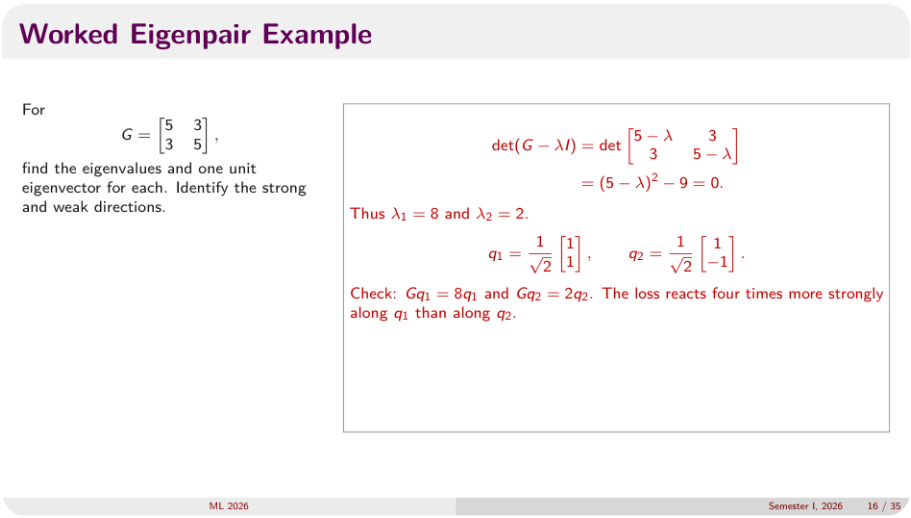
Slide 15 — Left: a general vector gets turned. Right: an eigenvector only gets scaled.

A nonzero q is an eigenvector of a square G when $Gq = \lambda q$.

- **General direction** u: Gu points somewhere else (direction changes).
- **Eigen-direction** q: $Gq = \lambda q$ stays on the **same line**, only scaled.

Example with $G = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$: $u = (1,0) \mapsto Gu = (5,3)$, turned. $q = (1,1) \mapsto (8,8)$, same line, ×8.

Slide 16 Worked Eigenpair Example



Slide 16 — Worked det(G − λI) = 0 → λ = 8, 2.

Find the eigenvalues and unit eigenvectors of $G = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$; identify the strong and weak directions.

Characteristic equation:

$$\det(G - \lambda I) = \det \begin{bmatrix} 5-\lambda & 3 \\ 3 & 5-\lambda \end{bmatrix} = (5-\lambda)^2 - 9 = 0 \;\Rightarrow\; 5 - \lambda = \pm 3 \;\Rightarrow\; \lambda_1 = 8,\ \lambda_2 = 2.$$

(Check: $8 + 2 = 10 =$ trace ✓; $8 \times 2 = 16 = 25 - 9 =$ det ✓.)

Eigenvectors. $\lambda = 8$: $(G - 8I)q = \begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix} q = 0 \Rightarrow q_a = q_b$. $\lambda = 2$: $\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} q = 0 \Rightarrow q_a = -q_b$.

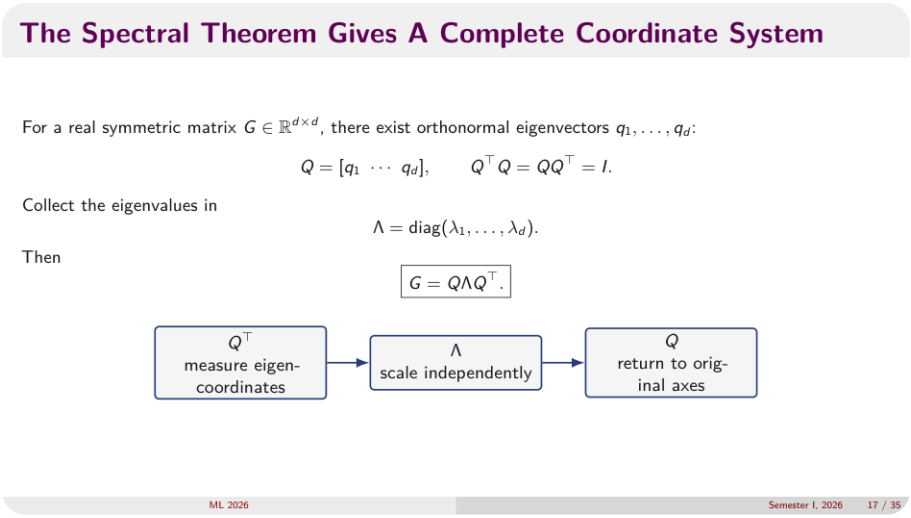
Normalise:

$$q_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad q_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Check: $Gq_1 = \frac{1}{\sqrt{2}}(8,8) = 8q_1$ ✓, $Gq_2 = \frac{1}{\sqrt{2}}(2,-2) = 2q_2$ ✓, $q_1^\top q_2 = \frac{1}{2}(1-1) = 0$ ✓.

Strong direction q_1 ($\lambda = 8$), **weak** direction q_2 ($\lambda = 2$). The loss reacts **four times more strongly** along q_1 than along q_2.

Slide 17 The Spectral Theorem Gives A Complete Coordinate System



Slide 17 — $Q^\top$ rotates in, Λ scales each axis, Q rotates back.

For real symmetric $G \in \mathbb{R}^{d\times d}$ there exist orthonormal eigenvectors $q_1..q_d$:

$$Q = [q_1 \ \cdots \ q_d], \quad Q^\top Q = QQ^\top = I, \quad \Lambda = \mathrm{diag}(\lambda_1..\lambda_d), \quad G = Q\Lambda Q^\top.$$

$Q^\top$: measure eigen-coordinates → Λ: scale independently → Q: return to original axes. (A6)

Check on slide 16: $Q = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$, $\Lambda = \mathrm{diag}(8,2)$.

$$Q\Lambda Q^\top = \tfrac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}8 & 0\\ 0 & 2\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} = \tfrac{1}{2}\begin{bmatrix}8 & 2\\ 8 & -2\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} = \tfrac{1}{2}\begin{bmatrix}10 & 6\\ 6 & 10\end{bmatrix} = G \checkmark.$$

Or as a sum: $8q_1q_1^\top + 2q_2q_2^\top = 4\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix} + \begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix} = \begin{bmatrix}5 & 3\\ 3 & 5\end{bmatrix}$ ✓.

Why symmetric matters: $X^\top X$ and C are symmetric, so this always works for them. (Non-symmetric matrices may have complex eigenvalues or not enough eigenvectors.)

Slide 18 Change Of Basis: The Vector Stays, Its Coordinates Change

Q orthogonal. For a coefficient vector θ, its eigenbasis coordinates are

$$a = Q^\top \theta, \qquad a_j = q_j^\top \theta \ \ (\text{signed coordinate along } q_j).$$

Because $QQ^\top = I$: $\theta = Qa = a_1 q_1 + \cdots + a_d q_d$.

> **Summary:** changing basis does **not** physically rotate the coefficient vector. It changes the axes used to describe it. (A5 has a worked example.)

Interpretation for slide 14: $a_1 = q_1^\top \theta = (\theta_{\text{rain}} + \theta_{\text{wet}})/\sqrt{2}$ is the "combined effect" coordinate; $a_2 = (\theta_{\text{rain}} - \theta_{\text{wet}})/\sqrt{2}$ is the "difference" coordinate.

Slide 19 Rotate The Normal Equation Without Shorthand

Substitute $X^\top X = Q\Lambda Q^\top$ and left-multiply by $Q^\top$. Don't introduce a, b yet.

Line	Reason
$X^\top X\hat{\theta} = X^\top y$	normal equation
$Q\Lambda Q^\top \hat{\theta} = X^\top y$	spectral decomposition
$Q^\top Q\Lambda Q^\top \hat{\theta} = Q^\top X^\top y$	left-multiply **both sides** by $Q^\top$
$\Lambda(Q^\top \hat{\theta}) = Q^\top X^\top y$	$Q^\top Q = I$

Both sides are now written in eigenvector coordinates.

Slide 20 Now Define a And b : Nothing Appears From Nowhere

The previous line is $\Lambda \underbrace{(Q^\top \hat{\theta})}_{a} = \underbrace{Q^\top X^\top y}_{b}$. So

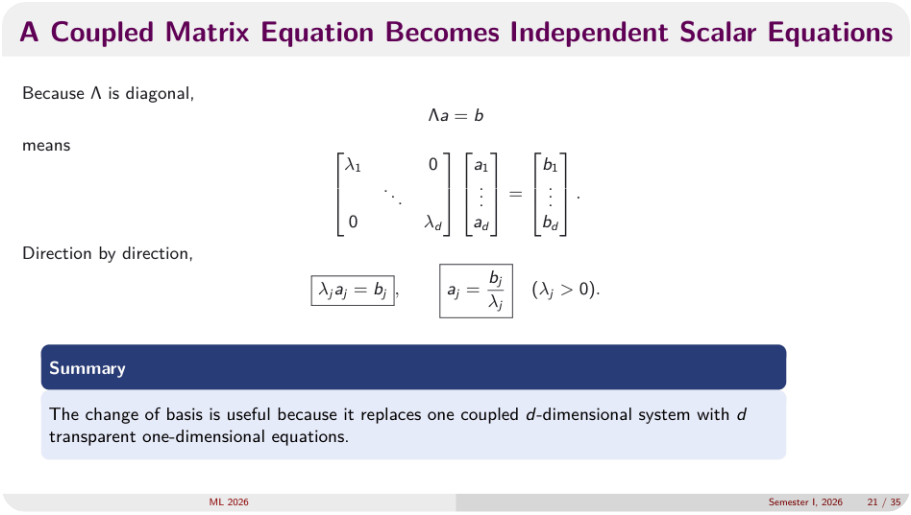
$$a := Q^\top \hat{\theta}, \qquad b := Q^\top X^\top y, \qquad \Lambda a = b.$$

- **Coefficient coordinate:** $a_j = q_j^\top \hat{\theta}$ (the fitted coefficients measured along q_j).

- **Response information in direction q_j:** $b_j = q_j^\top X^\top y = (Xq_j)^\top y$.

b_j measures the **alignment between the observed response y and the prediction pattern Xq_j** generated by moving coefficients along q_j. ($Xq_j \in \mathbb{R}^n$ is "what the predictions would do if you nudged θ along q_j"; dot it with y to see whether the data want that.)

Slide 21 A Coupled Matrix Equation Becomes Independent Scalar Equations



Slide 21 — A coupled 2×2 system becomes separate one-line equations.

Λ is diagonal, so $\Lambda a = b$ is

$$\begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_d \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_d \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_d \end{bmatrix} \quad \Longrightarrow \quad \lambda_j a_j = b_j, \qquad a_j = \frac{b_j}{\lambda_j} \ (\lambda_j > 0).$$

Then go back: $\hat{\theta} = Qa = \sum_j \frac{b_j}{\lambda_j} q_j$.

Summary: one coupled d-dimensional system → d transparent 1-D equations.

Worked example (slide 14 matrix). Suppose $X^\top y = (20,\ 19.8)$.

- $b_1 = q_1^\top X^\top y = \frac{39.8}{\sqrt{2}},\ b_2 = q_2^\top X^\top y = \frac{0.2}{\sqrt{2}}$.
- $a_1 = \frac{b_1}{19.8} = \frac{2.0101}{\sqrt{2}},\ a_2 = \frac{b_2}{0.2} = \frac{1}{\sqrt{2}}$.

- $\hat{\theta} = a_1 q_1 + a_2 q_2 = 1.00505\,(1,1) + 0.5\,(1,-1) = (1.505,\ 0.505)$.

Check: $10(1.505) + 9.8(0.505) = 15.05 + 4.95 = 20.0$ ✓; $9.8(1.505) + 10(0.505) = 14.75 + 5.05 = 19.8$ ✓.

Slide 22 Why A Small Eigenvalue Makes Coefficients Sensitive

Perturb $b \to b + \delta b$: $\Lambda(a + \delta a) = b + \delta b$. Subtract $\Lambda a = b$:

$$\Lambda\,\delta a = \delta b \quad \Rightarrow \quad \delta a_j = \frac{\delta b_j}{\lambda_j}.$$

- **Large λ_j:** a fixed data perturbation produces a small coefficient change.
- **Small λ_j:** dividing by a small number **amplifies** it.

Continue the slide-21 example. Change $X^\top y$ from $(20, 19.8)$ to $(20, 20)$ (a 1% nudge in one entry). Now $b_2 = 0$, so $a_2 = 0$, and $\hat{\theta} = \frac{40}{39.6}(1,1) = (1.010,\ 1.010)$.

	$X^\top y$	$\hat{\theta}$
before	$(20,\ 19.8)$	$(1.505,\ 0.505)$
after	$(20,\ 20)$	$(1.010,\ 1.010)$

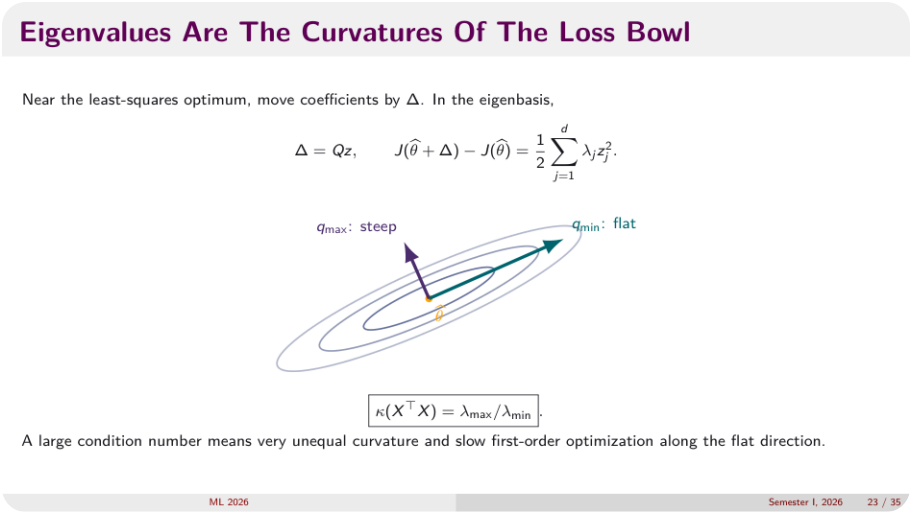
A 0.2 change in the data moved each coefficient by ~0.5: $\delta b_2 = -0.2/\sqrt{2}$ divided by $\lambda_2 = 0.2$ gives $\delta a_2 = -1/\sqrt{2}$. The sum $\theta_1 + \theta_2$ (the strong direction) barely changed: $2.010 \to 2.020$.

> **Summary:** predictions can remain similar, because the same weak direction also satisfies

$$\|Xq_j\|_2^2 = q_j^\top X^\top X q_j = \lambda_j.$$

Moving along a small-λ direction hardly changes $\hat{y}$. That's exactly **why** the data can't pin it down. (This is **multicollinearity**; ridge regression, L7, adds λ_{ridge} to every λ_j to stop the blow-up.)

Slide 23 Eigenvalues Are The Curvatures Of The Loss Bowl



Slide 23 — The bowl is steep along the big-λ direction, flat along the small-λ one.

Near the optimum, move by Δ. Because J is exactly quadratic and $\nabla J(\hat{\theta}) = 0$:

$$J(\hat{\theta}+\Delta) - J(\hat{\theta}) = \underbrace{\nabla J(\hat{\theta})^\top \Delta}_{0} + \tfrac{1}{2}\Delta^\top X^\top X \Delta.$$

In the eigenbasis, $\Delta = Qz$ (A6):

$$J(\hat{\theta}+\Delta) - J(\hat{\theta}) = \frac{1}{2}\sum_{j=1}^{d} \lambda_j z_j^2.$$

Each λ_j is the **curvature** along q_j. Contours are ellipses with axes along the q_j, and the half-length along q_j is $\propto 1/\sqrt{\lambda_j}$: $q_{\min}$ is the long **flat** axis, $q_{\max}$ the short **steep** one.

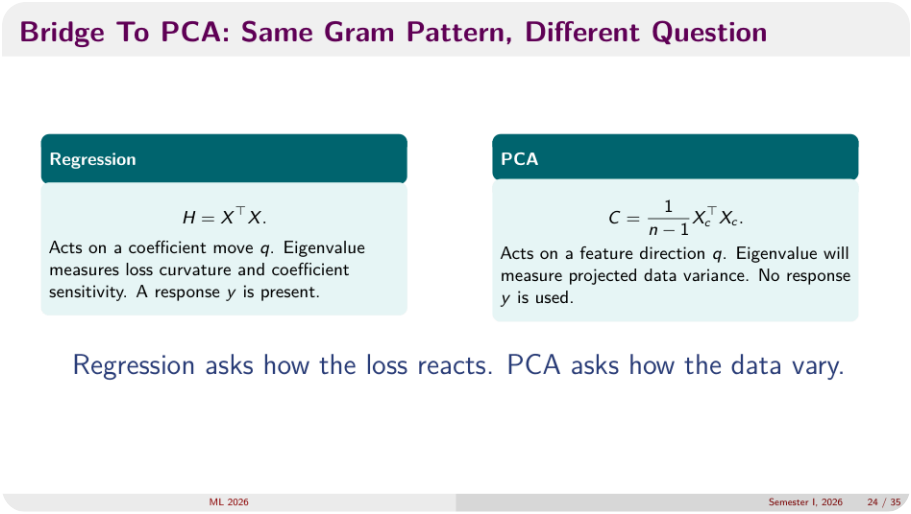
$$\kappa(X^\top X) = \frac{\lambda_{\max}}{\lambda_{\min}}.$$

A large condition number means very unequal curvature and **slow first-order optimisation** (gradient descent) along the flat direction: the step size must be small enough for the steep direction ($< 2/\lambda_{\max}$), so progress along the flat one is tiny. (A7)

Expanding the derivation. $J(\hat{\theta}+\Delta) = \frac{1}{2}\|\hat{r} - X\Delta\|^2 = \frac{1}{2}\|\hat{r}\|^2 - \hat{r}^\top X\Delta + \frac{1}{2}\|X\Delta\|^2$, and $\hat{r}^\top X = 0$ (slide 10).

Slide 24

Bridge To PCA: Same Gram Pattern, Different Question

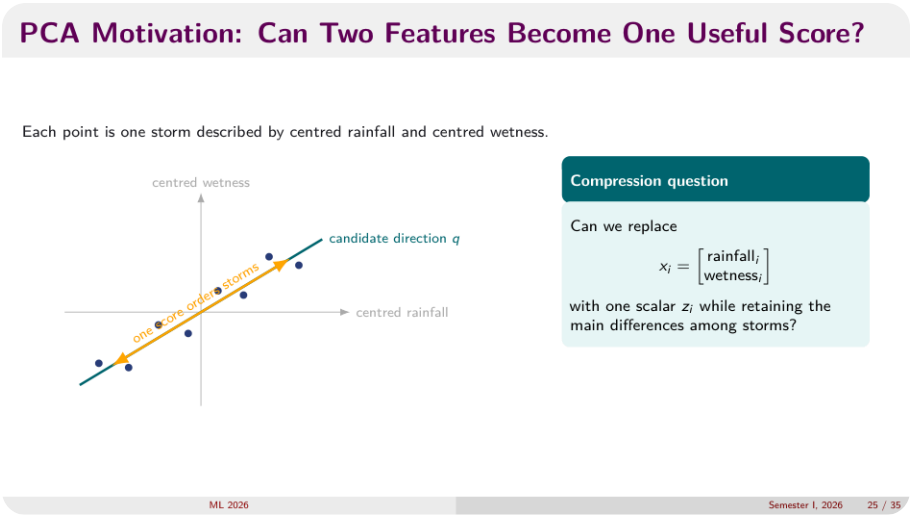


Slide 24 — Same Gram pattern: regression asks how the loss reacts; PCA asks how the data vary.

	Regression	PCA
matrix	$H = X^\top X$	$C = \frac{1}{n-1} X_c^\top X_c$
acts on	a coefficient move q	a feature direction q
eigenvalue measures	loss curvature, coefficient sensitivity	projected data variance
uses a response y?	yes	**no** (unsupervised)

Regression asks how the loss reacts. PCA asks how the data vary.

Slide 25 PCA Motivation: Can Two Features Become One Useful Score?



Slide 25 — The cloud is long along one diagonal; one score along it captures most differences.

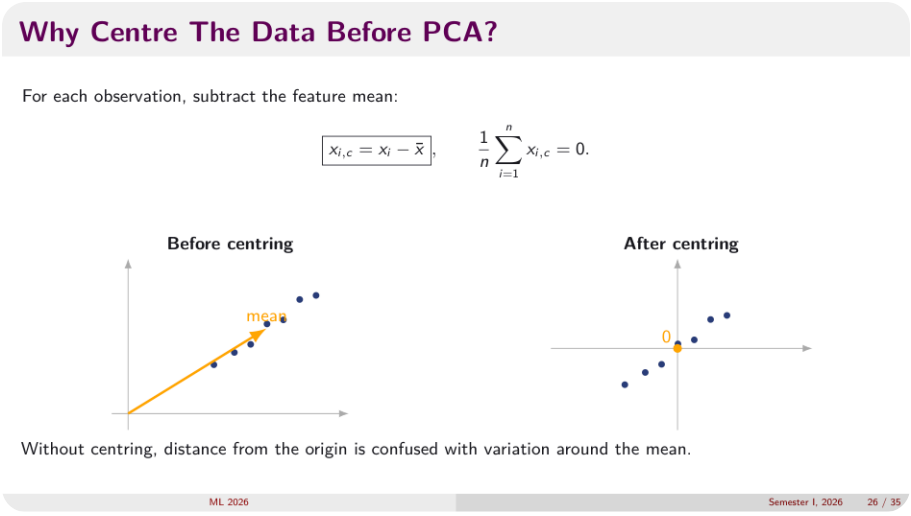
Each point is one storm described by centred rainfall and centred wetness. The cloud is elongated along a diagonal. A candidate direction q through the cloud gives each storm one number (its position along q), and that one score **orders the storms**.

Compression question: can we replace $x_i = \begin{bmatrix} \text{rainfall}_i \\ \text{wetness}_i \end{bmatrix}$ with one scalar z_i while **retaining the main differences among storms**?

(Our toy storms: points $(2, 2), (1, -1), (-1, 1), (-2, -2)$, spread most along the $(1, 1)$ diagonal.)

Slide 26

Why Centre The Data Before PCA?



Slide 26 — Before/after centring: subtract the mean so directions pass through the cloud.

For each observation subtract the feature mean:

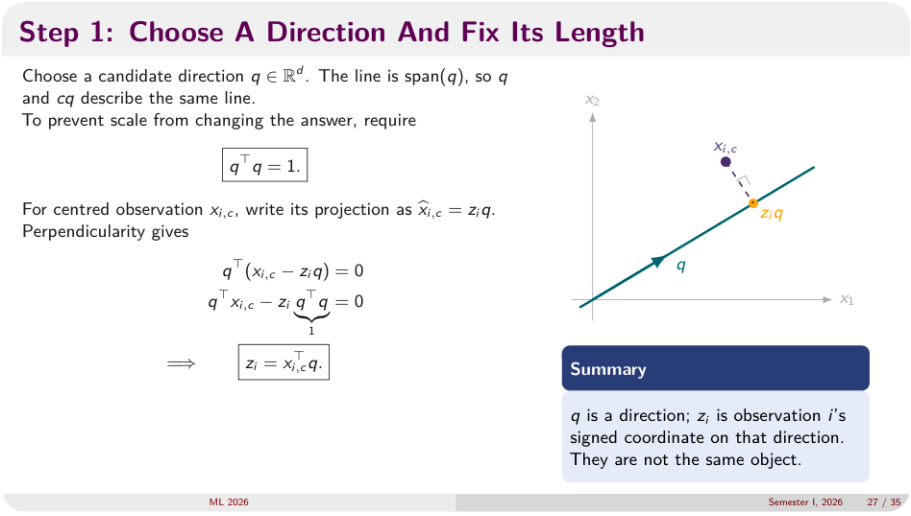
$$x_{i,c} = x_i - \bar{x}, \qquad \frac{1}{n}\sum_{i=1}^{n} x_{i,c} = 0.$$

Before centring the cloud sits around the mean; after, around 0.

Without centring, distance from the origin is confused with variation around the mean. (A8) The direction "towards the cloud's centre" would look like a big-variance direction even if the storms were nearly identical.

Example: raw storms $(52, 32), (51, 29), (49, 31), (48, 28)$. Mean $(50, 30)$. Subtract → our toy table ✓.

Slide 27 Step 1: Choose A Direction And Fix Its Length



Slide 27 — A unit direction q and a point's shadow $z_i = x_i^\top q$ on it.

Choose a candidate $q \in \mathbb{R}^d$. The line is $\mathrm{span}(q)$; q and cq describe the **same line**. To stop scale from changing the answer, require

$$q^\top q = 1.$$

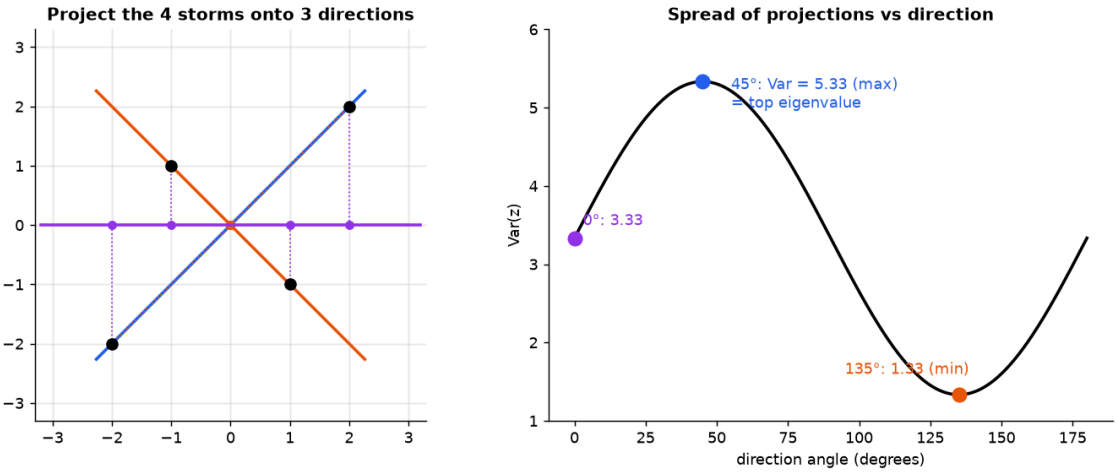
Projection of $x_{i,c}$ onto the line is $\hat{x}_{i,c} = z_i q$. Perpendicularity (A9):

$$q^\top (x_{i,c} - z_i q) = 0 \;\Rightarrow\; q^\top x_{i,c} - z_i \underbrace{q^\top q}_{1} = 0 \;\Rightarrow\; z_i = x_{i,c}^\top q.$$

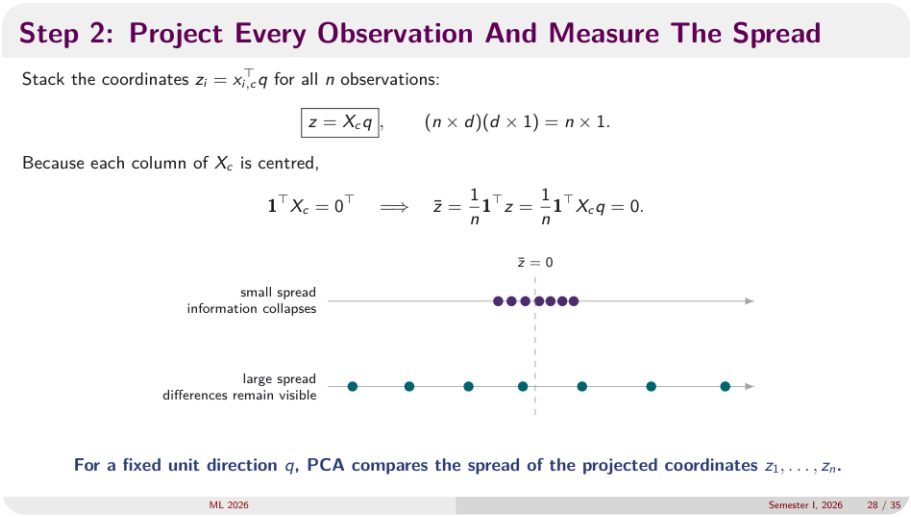
> **Summary:** q is a **direction**; z_i is observation i's **signed coordinate** on that direction. Not the same object.

Toy storms, $q = \frac{1}{\sqrt{2}}(1,1)$: storm 1 $z_1 = \frac{2+2}{\sqrt{2}} = 2\sqrt{2} \approx 2.83$; storm 2 $z_2 = \frac{1-1}{\sqrt{2}} = 0$.

Slide 28 Step 2: Project Every Observation And Measure The Spread



Slide 28's table as a picture: spin the direction, measure the spread; it peaks at 45° (5.33), the top eigenvalue.



Slide 28 — Bunched shadows (small spread) vs spread-out shadows (large spread).

Stack all coordinates:

$$z = X_c q, \qquad (n \times d)(d \times 1) = n \times 1.$$

Because each column of X_c is centred,

$$\mathbf{1}^\top X_c = 0^\top \;\Rightarrow\; \bar{z} = \tfrac{1}{n}\mathbf{1}^\top z = \tfrac{1}{n}\mathbf{1}^\top X_c q = 0.$$

(Centred data ⇒ centred scores, for **any** q.)

Small spread of z	Large spread of z
information collapses: storms land on top of each other	differences remain visible

For a fixed unit q, PCA compares the spread of $z_1..z_n$.

Toy storms, two candidates:

q	$z = X_c q$	$\mathrm{Var}(z) = \frac{z^\top z}{3}$
$\frac{1}{\sqrt{2}}(1,1)$	$\frac{1}{\sqrt{2}}(4,0,0,-4) = (2.83,0,0,-2.83)$	$\frac{8+8}{3} = 5.33$
$\frac{1}{\sqrt{2}}(1,-1)$	$\frac{1}{\sqrt{2}}(0,2,-2,0) = (0,1.41,-1.41,0)$	$\frac{2+2}{3} = 1.33$
$(1,0)$ (rainfall only)	$(2,1,-1,-2)$	$\frac{10}{3} = 3.33$

The diagonal $(1,1)$ keeps far more spread.

Slide 29

Step 3: The Objective And Constraint Now Fit Together

Since $\bar{z} = 0$:

$$\mathrm{Var}(z) = \frac{1}{n-1}\sum_i z_i^2 = \frac{1}{n-1} z^\top z = \frac{1}{n-1}(X_c q)^\top (X_c q) = q^\top \underbrace{\left(\frac{1}{n-1} X_c^\top X_c\right)}_{C} q.$$

Define the **feature covariance matrix** $C = \frac{1}{n-1} X_c^\top X_c$. The first PCA direction is

$$\boxed{q_1 = \arg\max_{q \in \mathbb{R}^d} q^\top C q \quad \text{subject to} \quad q^\top q = 1.}$$

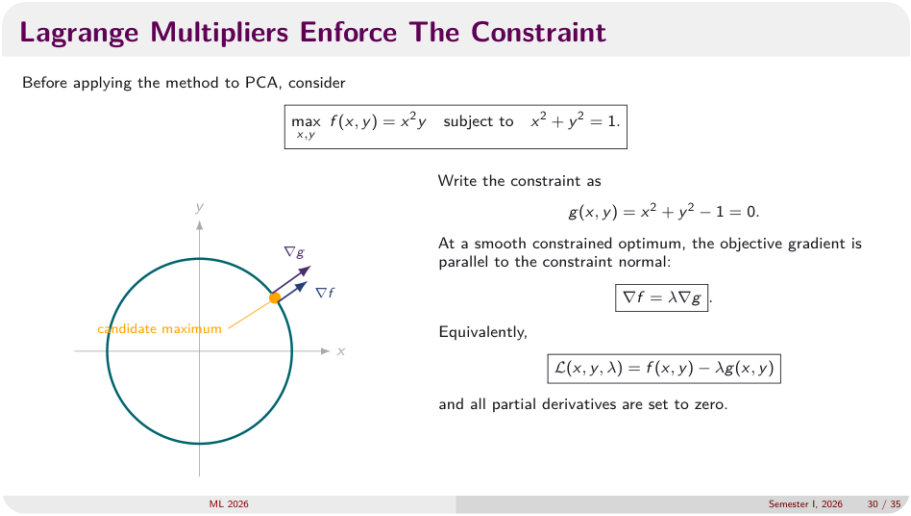
- **Objective** $q^\top C q$: retain the largest projected variance.
- **Constraint** $q^\top q = 1$: prevent scale from manufacturing variance (without it, $q = 10 q_1$ gives 100× the "variance", and the max is ∞).

Toy storms: $X_c^\top X_c = \begin{bmatrix} 4+1+1+4 & 4-1-1+4 \\ 6 & 4+1+1+4 \end{bmatrix} = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}$, so

$$C = \frac{1}{3}\begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix} = \begin{bmatrix} 3.33 & 2 \\ 2 & 3.33 \end{bmatrix}.$$

Each feature has variance 3.33; covariance 2 (positively correlated, correlation $2/3.33 = 0.6$). Check slide 28: $q = \frac{1}{\sqrt{2}}(1,1)$ gives $q^\top C q = \frac{1}{2}(3.33 + 2 + 2 + 3.33) = 5.33$ ✓.

Slide 30 Lagrange Multipliers Enforce The Constraint



Slide 30 — On the circle, the best point is where ∇f is parallel to ∇g.

Warm-up before PCA:

$$\max_{x,y} f(x,y) = x^2 y \quad \text{subject to} \quad x^2 + y^2 = 1.$$

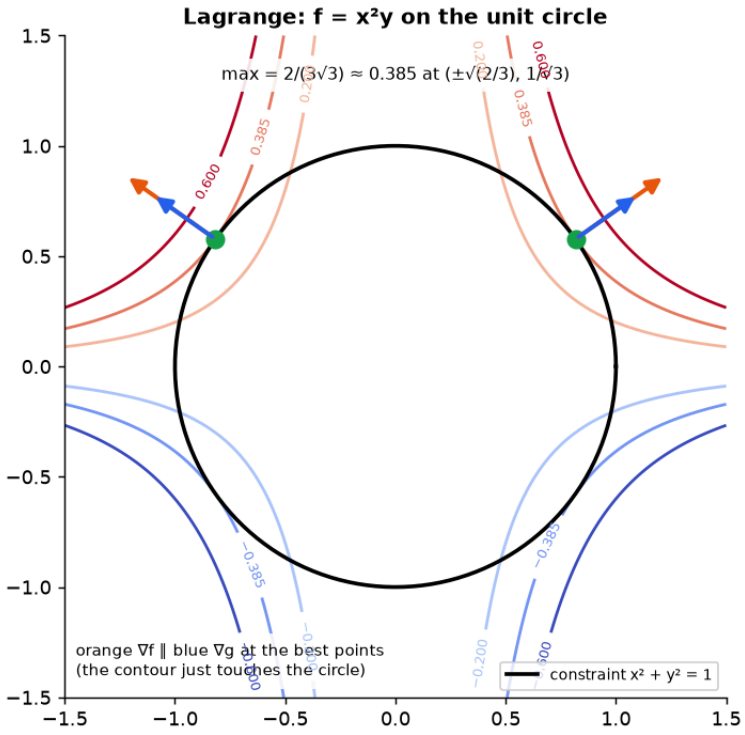
Constraint as $g(x,y) = x^2 + y^2 - 1 = 0$. At a smooth constrained optimum, the objective gradient is **parallel to the constraint normal**:

$$\nabla f = \lambda \nabla g.$$

Equivalently form $\mathcal{L}(x,y,\lambda) = f(x,y) - \lambda g(x,y)$ and set **all** partials to zero. (A10)

Picture: walking around the unit circle, at the best point ∇f points straight out of the circle (along ∇g). If it had any component along the circle, you could walk that way and do better.

Slide 31 Solve $\max x^2y$ On The Unit Circle Step By Step



The best points are where a contour of x²y just touches the circle; there ∇f and ∇g line up.

Solve $\max x^2y$ On The Unit Circle Step By Step

For

$$\mathcal{L} = x^2y - \lambda(x^2 + y^2 - 1),$$

derive the stationary equations, separate the $x = 0$ and $x \neq 0$ cases, and identify the global maximum.

$$\frac{\partial \mathcal{L}}{\partial x} = 2xy - 2\lambda x = 0,$$
$$\frac{\partial \mathcal{L}}{\partial y} = x^2 - 2\lambda y = 0,$$
$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(x^2 + y^2 - 1) = 0.$$

If $x = 0$, then $y = \pm 1$ and $f = 0$.
If $x \neq 0$, the first equation gives $\lambda = y$. The second gives $x^2 = 2y^2$. With $x^2 + y^2 = 1$,

$$3y^2 = 1, \quad y = \pm\frac{1}{\sqrt{3}}, \quad x = \pm\sqrt{\frac{2}{3}}.$$

The global maximum is

$$\boxed{f_{\max} = \frac{2}{3\sqrt{3}}}$$

at $y = 1/\sqrt{3}$ and $x = \pm\sqrt{2/3}$.

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Slide 31 — The worked max of x²y on the unit circle.

$$\mathcal{L} = x^2y - \lambda(x^2 + y^2 - 1).$$

Stationary equations:

$$\frac{\partial \mathcal{L}}{\partial x} = 2xy - 2\lambda x = 0, \qquad \frac{\partial \mathcal{L}}{\partial y} = x^2 - 2\lambda y = 0, \qquad \frac{\partial \mathcal{L}}{\partial \lambda} = -(x^2 + y^2 - 1) = 0.$$

The first factorises: $2x(y - \lambda) = 0$, so **either** $x = 0$ **or** $\lambda = y$. (Don't divide by x without splitting cases.)

Case $x = 0$**:** constraint $\Rightarrow y = \pm 1$; second equation $\Rightarrow 0 = 2\lambda y \Rightarrow \lambda = 0$. Then $f = 0$.

Case $x \neq 0$**:** $\lambda = y$. Second equation: $x^2 = 2\lambda y = 2y^2$. Constraint: $2y^2 + y^2 = 1 \Rightarrow y^2 = \frac{1}{3}$:

$$y = \pm\frac{1}{\sqrt{3}}, \qquad x = \pm\sqrt{\tfrac{2}{3}}.$$

Evaluate every candidate:

x	y	$f = x^2 y$
0	± 1	0
$\pm\sqrt{2/3}$	$+1/\sqrt{3}$	$\frac{2}{3} \cdot \frac{1}{\sqrt{3}} = \frac{2}{3\sqrt{3}} \approx 0.385$
$\pm\sqrt{2/3}$	$-1/\sqrt{3}$	$-\frac{2}{3\sqrt{3}} \approx -0.385$

Global maximum $f_{\max} = \dfrac{2}{3\sqrt{3}}$ at $y = 1/\sqrt{3}$, $x = \pm\sqrt{2/3}$. (The negative ones are the global minima.)

Check $\nabla f \parallel \nabla g$ at $(\sqrt{2/3}, 1/\sqrt{3})$: $\nabla f = (2xy, x^2) = (2\sqrt{2}/3,\ 2/3)$; $\nabla g = (2x, 2y) = (2\sqrt{2/3},\ 2/\sqrt{3})$. Ratio in both components: $\lambda = \frac{1}{\sqrt{3}} = y$ ✓.

Slide 32 The PCA Constraint Produces An Eigenvalue Equation

Same method on $\max_q q^\top C q$ s.t. $q^\top q = 1$:

$$\mathcal{L}(q, \lambda) = q^\top C q - \lambda(q^\top q - 1).$$

Because C is symmetric, $\nabla_q(q^\top C q) = 2Cq$ and $\nabla_q(q^\top q) = 2q$ (L4 A9):

$$\nabla_q \mathcal{L} = 2Cq - 2\lambda q = 0 \quad \Longrightarrow \quad \boxed{Cq = \lambda q}.$$

Every stationary direction is an eigenvector of C, and the multiplier is its eigenvalue.

Which one wins? At a unit eigenvector q_j:

$$q_j^\top C q_j = q_j^\top (\lambda_j q_j) = \lambda_j \, q_j^\top q_j = \lambda_j.$$

The **variance captured equals the eigenvalue**. So maximising projected variance means

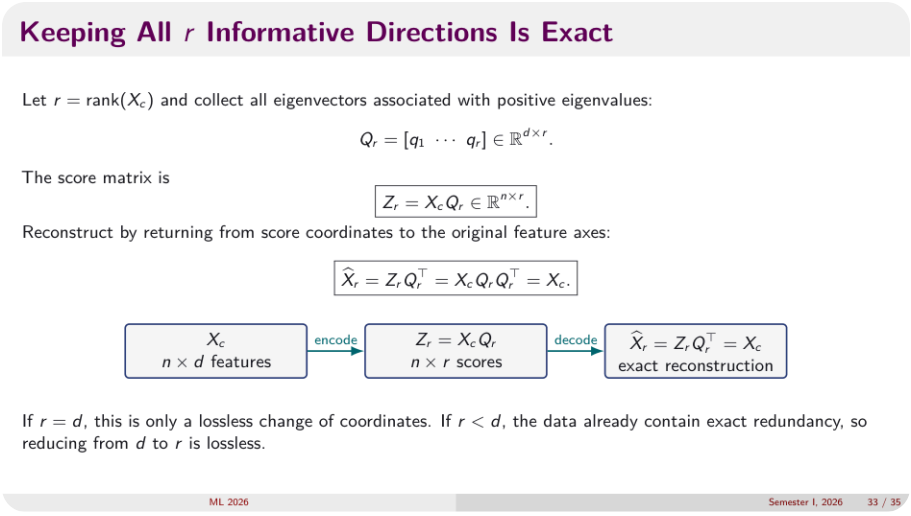
$$q^* = q_1, \qquad \lambda_1 = \max_j \lambda_j.$$

(The Rayleigh bound, A6, confirms no non-eigenvector does better.)

The candidate was named q; after optimisation, the winning direction is the top eigenvector q_1.

Toy storms: $C = \frac{1}{3}\begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}$: eigenvalues $\frac{10 \pm 6}{3} = \frac{16}{3}, \frac{4}{3}$ $(5.33, 1.33)$, eigenvectors $\frac{1}{\sqrt{2}}(1,1)$, $\frac{1}{\sqrt{2}}(1,-1)$. Exactly the two variances in the slide-28 table: PC1 = "storm intensity" (rain and wetness together), PC2 = "rain vs wetness contrast".

Slide 33 Keeping All r Informative Directions Is Exact



Slide 33 — Keep all r directions: exact reconstruction, just new coordinates.

Let $r = \operatorname{rank}(X_c)$ and collect the eigenvectors with **positive** eigenvalues: $Q_r = [q_1 \ \cdots \ q_r] \in \mathbb{R}^{d \times r}$.

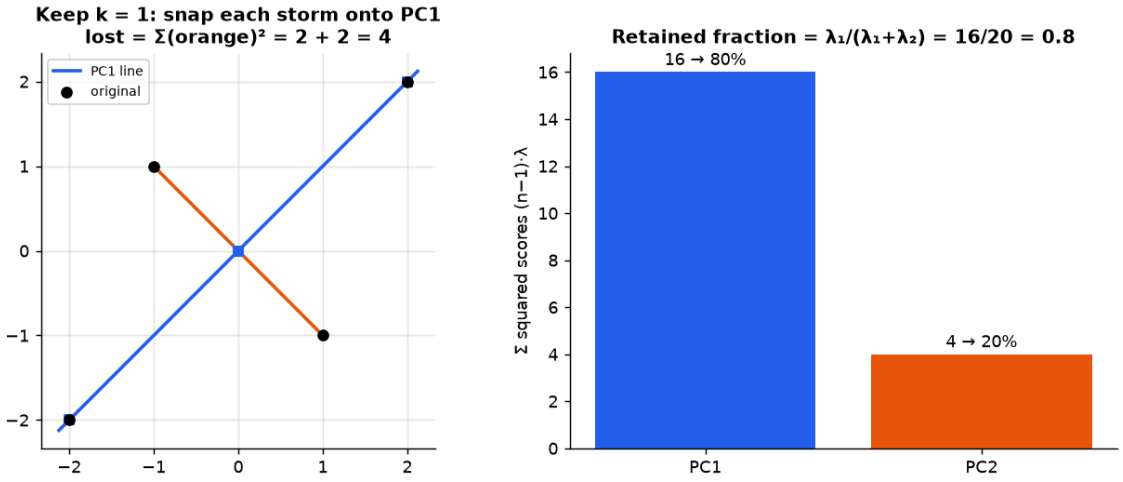
- **Encode** (scores): $Z_r = X_c Q_r \in \mathbb{R}^{n \times r}$.
- **Decode** (back to feature axes): $\hat{X}_r = Z_r Q_r^\top = X_c Q_r Q_r^\top = X_c$.

Why exactly X_c: the dropped directions $q_{r+1}..q_d$ have $\lambda_j = 0$, so $\|X_c q_j\|^2 = (n-1)\lambda_j = 0$, i.e. $X_c q_j = 0$. Then $X_c = X_c \sum_{j=1}^{d} q_j q_j^\top = X_c Q_r Q_r^\top + 0$.

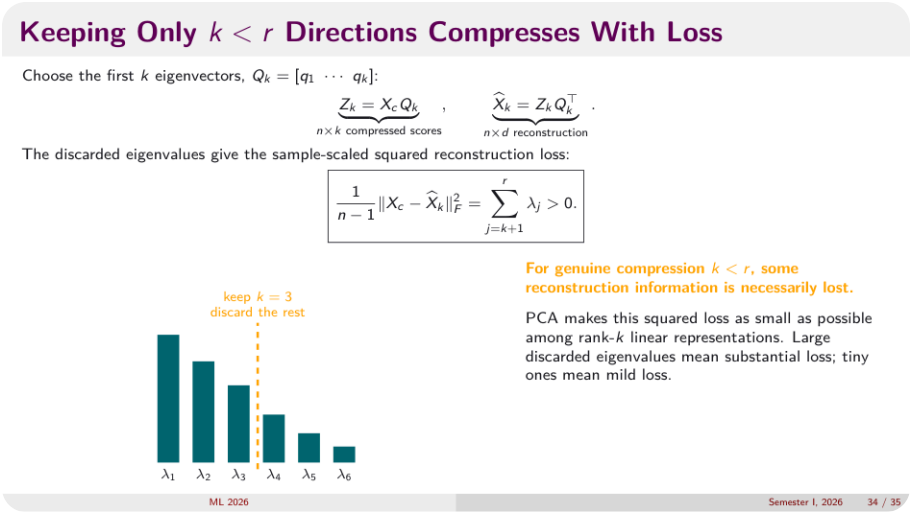
- If $r = d$: just a **lossless change of coordinates** (a rotation).
- If $r < d$: the data already contain **exact redundancy** (some feature is a combination of others), so going from d to r numbers is lossless.

Toy storms: $r = d = 2$ (both eigenvalues positive). $Z_2 = X_c Q$ = rows $(2.83, 0), (0, 1.41), (0, -1.41), (-2.83, 0)$: the same four storms in rotated axes.

Slide 34 Keeping Only $k < r$ Directions Compresses With Loss



Keep 1 of 2 directions: each storm snaps onto PC1; the lost part is the orange gaps (4 of 20, so 80% kept).



Slide 34 — Keep k < r: the bar chart shows how much each dropped λ costs.

Choose the first k eigenvectors $Q_k = [q_1 \cdots q_k]$:

$$\underbrace{Z_k = X_c Q_k}_{n \times k \text{ compressed scores}} \quad , \qquad \underbrace{\hat{X}_k = Z_k Q_k^\top}_{n \times d \text{ reconstruction}} \ .$$

The discarded eigenvalues give the sample-scaled squared reconstruction loss (proof in A11):

$$\frac{1}{n-1} \| X_c - \hat{X}_k \|_F^2 = \sum_{j=k+1}^{r} \lambda_j > 0.$$

Bar-chart picture: $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_6$; keep $k = 3$, discard the rest; the discarded bars' total height is the loss.

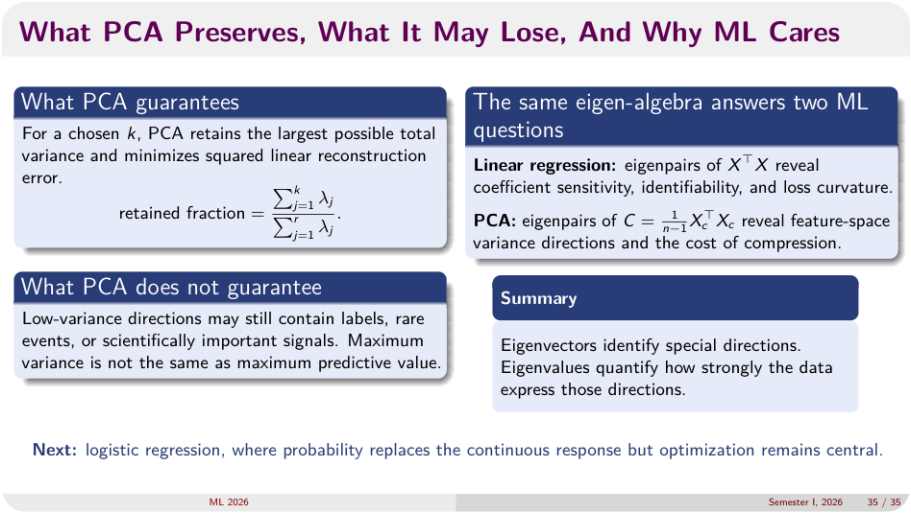
- For genuine compression ($k < r$) some information is **necessarily** lost.
- PCA makes this squared loss **as small as possible** among all rank-k linear representations (you drop the smallest eigenvalues, so the sum you lose is the smallest possible).
- Large discarded eigenvalues = substantial loss; tiny ones = mild loss.

Toy storms, $k = 1$**:** $z = (2.83, 0, 0, -2.83)$, $\hat{X}_1 = z q_1^\top$:

storm	$x_{i,c}$	$\hat{x}_i = z_i q_1$	error
1	$(2, 2)$	$2.83 \cdot \frac{1}{\sqrt{2}}(1, 1) = (2, 2)$	$(0, 0)$
2	$(1, -1)$	$(0, 0)$	$(1, -1)$
3	$(-1, 1)$	$(0, 0)$	$(-1, 1)$
4	$(-2, -2)$	$(-2, -2)$	$(0, 0)$

$\| X_c - \hat{X}_1 \|_F^2 = 1 + 1 + 1 + 1 = 4$; divided by $n - 1 = 3$: $\frac{4}{3} = \lambda_2$ ✓.

Slide 35 What PCA Preserves, What It May Lose, And Why ML Cares



Slide 35 — What PCA guarantees, what it doesn't, and the shared eigen-algebra.

What PCA guarantees. For a chosen k, PCA retains the **largest possible total variance** and **minimises squared linear reconstruction error**.

$$\text{retained fraction} = \frac{\sum_{j=1}^{k} \lambda_j}{\sum_{j=1}^{r} \lambda_j}.$$

(Toy storms, $k = 1$: $\frac{16/3}{20/3} = 80\%$. The denominator is also $\text{trace}(C)$ = total variance $= 3.33 + 3.33$ ✓.)

What PCA does not guarantee. Low-variance directions may still contain **labels, rare events, or scientifically important signals**. Maximum variance ≠ maximum predictive value. (PCA never looks at y. E.g. the small "rain vs wetness contrast" might be exactly what separates flash floods from slow ones.)

Careful. PCA is also **scale-sensitive**: a feature in mm has bigger variance than the same feature in m. Standardise features first if units differ.

The same eigen-algebra answers two ML questions:

	Matrix	Eigenpairs reveal
linear regression	$X^\top X$	coefficient sensitivity, identifiability, loss curvature
PCA	$C = \frac{1}{n-1} X_c^\top X_c$	feature-space variance directions, the cost of compression

> **Summary:** eigenvectors identify **special directions**; eigenvalues quantify **how strongly the data express** them.

Next: logistic regression, where probability replaces the continuous response but optimisation remains central.

Part C Practice

1. For X with columns $\mathbf{1} = (1,1,1)$ and $x = (1,2,3)$, write $X^\top X$. Interpret each entry as a Gram-matrix dot product.
2. Using the componentwise formula $\partial J/\partial\theta_j = \sum_i x_{ij}(x_i^\top\theta - y_i)$, compute ∇J at $\theta = (0,1)$ for the data of Q1 with $y = (2,2,4)$.
3. Find the eigenvalues and unit eigenvectors of $\begin{bmatrix}4 & 1\\ 1 & 4\end{bmatrix}$. Verify $G = Q\Lambda Q^\top$.
4. Find the eigenvalues of $\begin{bmatrix}2 & 1\\ 1 & 3\end{bmatrix}$ with the trace/determinant shortcut. What is the condition number?
5. Show that eigenvectors of a symmetric matrix with different eigenvalues are orthogonal. (Hint: compute $q_1^\top G q_2$ two ways.)
6. With $q_1 = \frac{1}{\sqrt{2}}(1,1)$, $q_2 = \frac{1}{\sqrt{2}}(1,-1)$, find the eigen-coordinates a of $\theta = (5,-1)$ and rebuild θ from them.
7. $X^\top X = \begin{bmatrix}5 & 3\\ 3 & 5\end{bmatrix}$, $X^\top y = (8,4)$. Solve the normal equation **through the eigenbasis** ($b = Q^\top X^\top y$, $a_j = b_j/\lambda_j$, $\hat\theta = Qa$). Check by direct solve.
8. Show $\|Xq_j\|^2 = \lambda_j$ for a unit eigenvector of $X^\top X$. What does $\lambda_j = 0$ mean for predictions?
9. Why can't we write $\hat\theta = (X^\top X)^{-1}X^\top y$ if two predictor columns are identical? Does the normal equation still have a solution?
10. Maximise $f(x,y) = xy$ on $x^2 + y^2 = 1$ with Lagrange multipliers. Which λ goes with the maximum?
11. Points $(3,1), (1,3), (-1,-3), (-3,-1)$ (already centred). Compute C, its eigenpairs, PC1 scores, and the retained-variance fraction for $k = 1$.
12. Why must PCA constrain $q^\top q = 1$? What happens to $\max q^\top C q$ without it?
13. For Q11, compute the $k = 1$ reconstruction error $\frac{1}{n-1}\|X_c - \hat X_1\|_F^2$ directly and compare with λ_2.
14. A colleague drops all but the top 2 PCs before training a classifier and accuracy collapses. Give one reason that could happen.

1. $X^\top X = \begin{bmatrix}3 & 6\\ 6 & 14\end{bmatrix}$: $\mathbf{1}^\top\mathbf{1} = 3 = n$, $\mathbf{1}^\top x = 6 = \sum x_i$, $x^\top x = 14 = \sum x_i^2$.
2. $X\theta = (1,2,3)$, $X\theta - y = (-1,0,-1)$. $\partial/\partial\theta_0 = -1 + 0 - 1 = -2$; $\partial/\partial\theta_1 = 1(-1) + 2(0) + 3(-1) = -4$. $\nabla J = (-2,-4)$.
3. $\lambda = 4 \pm 1 = 5, 3$; $q_1 = \frac{1}{\sqrt{2}}(1,1)$, $q_2 = \frac{1}{\sqrt{2}}(1,-1)$. $5q_1q_1^\top + 3q_2q_2^\top = \frac{5}{2}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix} + \frac{3}{2}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix} = \begin{bmatrix}4 & 1\\ 1 & 4\end{bmatrix}$ ✓.

4. $T = 5$, $D = 5$: $\lambda = 2.5 \pm \sqrt{6.25 - 5} = 2.5 \pm 1.118 = 3.618,\ 1.382$. $\kappa = 3.618/1.382 \approx 2.62$.
5. $q_1^\top G q_2 = q_1^\top (\lambda_2 q_2) = \lambda_2 q_1^\top q_2$. Also $q_1^\top G q_2 = (G^\top q_1)^\top q_2 = (G q_1)^\top q_2 = \lambda_1 q_1^\top q_2$ (uses $G^\top = G$). Subtract: $(\lambda_1 - \lambda_2) q_1^\top q_2 = 0$, and $\lambda_1 \neq \lambda_2 \Rightarrow q_1^\top q_2 = 0$.
6. $a_1 = \frac{4}{\sqrt{2}} = 2\sqrt{2}$, $a_2 = \frac{6}{\sqrt{2}} = 3\sqrt{2}$. $2\sqrt{2} q_1 + 3\sqrt{2} q_2 = (2, 2) + (3, -3) = (5, -1)$ ✓.
7. $b_1 = \frac{12}{\sqrt{2}}$, $b_2 = \frac{4}{\sqrt{2}}$. $a_1 = \frac{12}{8\sqrt{2}} = \frac{1.5}{\sqrt{2}}$, $a_2 = \frac{4}{2\sqrt{2}} = \frac{2}{\sqrt{2}}$. $\hat{\theta} = 0.75(1, 1) + 1(1, -1) = (1.75, -0.25)$. Direct: $5(1.75) + 3(-0.25) = 8$ ✓, $3(1.75) + 5(-0.25) = 4$ ✓.
8. $\|X q_j\|^2 = q_j^\top X^\top X q_j = q_j^\top \lambda_j q_j = \lambda_j$. $\lambda_j = 0 \Rightarrow X q_j = 0$: moving coefficients along q_j changes **no** prediction, so the data say nothing about that direction.
9. Identical columns $\Rightarrow Xv = 0$ for $v = (1, -1, 0, \dots) \Rightarrow X^\top X$ singular (eigenvalue 0) $\Rightarrow$ no inverse. The normal equation still has solutions (always: $X^\top y$ lies in $\mathrm{Col}(X^\top) = \mathrm{Col}(X^\top X)$), infinitely many.
10. $\nabla f = (y, x) = \lambda(2x, 2y) \Rightarrow y = 2\lambda x$, $x = 2\lambda y \Rightarrow x = 4\lambda^2 x$. $x = 0$ forces $y = 0$, violating the constraint, so $\lambda = \pm\frac{1}{2}$. $\lambda = \frac{1}{2}$: $x = y = \pm\frac{1}{\sqrt{2}}$, $f = \frac{1}{2}$ (max). $\lambda = -\frac{1}{2}$: $x = -y$, $f = -\frac{1}{2}$ (min). The multiplier equals the optimal value here (like PCA, where λ = variance).
11. $X_c^\top X_c = \begin{bmatrix} 9+1+1+9 & 3+3+3+3 \\ 12 & 1+9+9+1 \end{bmatrix} = \begin{bmatrix} 20 & 12 \\ 12 & 20 \end{bmatrix}$, $C = \frac{1}{3}\begin{bmatrix} 20 & 12 \\ 12 & 20 \end{bmatrix}$. $\lambda = \frac{32}{3}, \frac{8}{3}$; $q_1 = \frac{1}{\sqrt{2}}(1, 1)$, $q_2 = \frac{1}{\sqrt{2}}(1, -1)$. Scores $z = \frac{1}{\sqrt{2}}(4, 4, -4, -4) = (2.83, 2.83, -2.83, -2.83)$. Retained $= \frac{32}{40} = 80\%$.
12. $q^\top C q$ scales as c^2 when $q \to cq$, so without a constraint the max is $+\infty$ (just make q longer). The constraint compares directions, not lengths.
13. $\hat{x}_i = z_i q_1$: $(2, 2), (2, 2), (-2, -2), (-2, -2)$. Errors $(1, -1), (-1, 1), (1, -1), (-1, 1)$, each squared norm 2, total 8. $8/3 = \lambda_2$ ✓.
14. PCA ignores the labels: the class-separating signal may sit in a low-variance direction that got dropped (slide 35). Also unscaled features can let one large-unit feature dominate the top PCs.